

Module 2 — 100 Question Revision Bank

Absolute Block • Colour Light Signalling • AWS • TPWS • TW5 Procedures • Shunting • Cab Prep • Coupling

All questions and answers are sourced exclusively from the uploaded rule book material (GERT8000-S7, SS2, TW1, TW5, RS521, RS522) and the classroom Shunting presentation. Each answer is followed by its citation.

Section 1 — Absolute Block Working & Signal Layout

1. A block section may have up to how many home signals? If there are two, what is the furthest from the box called, and what is the nearest called?

Answer: Up to 4 home signals. The furthest from the box is the Outer Home; the nearest is the Inner Home.

GERT8000-S7 §1.1b

2. If two stop signals are mounted on the same post, how must the driver respond?

Answer: Obey both signals which apply to the movement of the train.

GERT8000-S7 §1.1b

3. A driver has stopped or nearly stopped at a semaphore signal at danger, which then clears. What must they be prepared to do, and what is the exception?

Answer: Be prepared to stop at the next stop signal worked by the same signal box.

Exception: does not apply to a signal controlling entrance to an Intermediate Block Signal (IBS) section.

GERT8000-S7 §1.6

4. A stop signal and a distant signal are both not showing correctly. How must each be treated?

Answer: The stop signal must be treated as at danger; the distant signal must be treated as at caution.

GERT8000-S7 §1.5

5. If a distant signal is showing OFF, what must also be true of its associated stop signals?

Answer: All associated stop signals must also be off (clear).

Semaphore Signals Revision Notes, §2

6. What is a fixed distant signal, and how must it always be treated?

Answer: A board-painted (non-moving) distant signal, sometimes positioned before a buffer stop or on the approach to a terminal area. It must always be treated as at caution.

Semaphore Signals Revision Notes, §2

7. What is an outer distant signal, and when is one provided?

Answer: A distant signal fitted ahead of the normal distant signal, provided where block sections are unusually long, to give an earlier warning.

Semaphore Signals Revision Notes, §2

8. Define "station limits" in absolute block working, and who is responsible for movements within them.

Answer: Station limits run from the first home signal up to and including the section signal. The signaller for that box is responsible for all movements within them.

Semaphore Signals Revision Notes, §3

9. What is the clearance point (overlap) beyond a home signal, and what does it protect against?

Answer: Track beyond a home signal which must be clear before the previous signal is returned to danger. It protects against a train slightly over-running after the signal clears.

Semaphore Signals Revision Notes, §3

10. What is the purpose of an Intermediate Block Signal (IBS), and how is it identified on the post?

Answer: An IBS divides a long block section into two shorter sections to increase line capacity. It is identified by a white square sign with a vertical black line on the signal post.

Semaphore Signals Revision Notes, §5

Section 2 — Colour Light Signalling

11. A signal has a plain black plate with white text showing its number. What type of signal is this?

Answer: A controlled signal — operated by the signaller.

RS521 §1.2

12. A signal plate has a white horizontal stripe across the top of a black plate. What does this identify?

Answer: An automatic signal — operated by the passage of trains. The signaller can place it at danger using a signal post replacement switch.

RS521 §1.2

13. A signal plate has the word "SEMI" printed above the plate number. What does this mean?

Answer: It is a semi-automatic signal — normally operated by the passage of trains but can also be controlled from a signal box or ground frame, or by a signal post replacement switch.

RS521 §1.2

14. A signal plate shows a white upward arrow and the letter "R". What does this identify?

Answer: A distant signal. A distant signal cannot show a stop aspect or indication.

RS521 §1.2

15. What type of signal is identified by the letters "BR" on its plate, and why is it provided?

Answer: A banner repeating signal — provided at signals with restricted sighting.

RS521 §5.7

16. A signal plate shows the letters "CA". What does this mean and what is the signal's function?

Answer: It is a co-acting signal — it mirrors the main signal to aid sighting from close range.

RS521 §5.7

17. In three-aspect signalling, what does a single yellow aspect mean?

Answer: Caution — be prepared to stop at the next signal.

RS521 §2.1

18. In four-aspect signalling, what does a double yellow aspect mean?

Answer: Preliminary caution — be prepared to find the next signal showing a single yellow.

RS521 §2.2

19. A colour light signal is not showing any aspect at all. What must the driver treat it as?

Answer: A signal at danger — stop.

GERT8000-S7 §1.5

20. A driver stops or nearly stops at a 2-aspect colour light signal at danger. The signal then clears to proceed. What must the driver be prepared to do?

Answer: Be prepared to stop at the next stop signal worked by the same signal box.

GERT8000-S7 §1.6

21. A junction indicator is illuminated showing a line of white lights angled to the LEFT of the signal. What does this tell the driver?

Answer: A diverging route to the left is set.

RS521 §2.3

22. What does a flashing yellow aspect mean?

Answer: Facing points at a junction ahead are set for a diverging route with a lower speed than the straight route.

RS521 §2.5

23. A position-light signal associated with a main aspect is unlit. What does this mean?

Answer: Obey the main signal.

RS521 §2.7

24. Can a driver proceed with a passenger train on the authority of a position-light signal alone?

Answer: No — not unless entering a permissive platform line on the authority of a position-light or subsidiary signal associated with a main aspect.

GERT8000-S7 §3.1

25. A banner repeating signal is in the ON position (horizontal). What does this tell the driver?

Answer: The signal it applies to is at danger or caution.

RS521 §5.7

Section 3 — AWS (Automatic Warning System)

26. What was the original purpose of AWS?

Answer: To provide the driver with an audible and visual indication of whether the distant signal was clear or at caution. Should the driver fail to respond to a warning indication, an emergency brake application will be initiated.

RS522 §1.1.2

27. How far from a signal is the AWS magnet normally positioned? When can it differ?

Answer: Normally 180 metres (approximately 200 yards). It may be at a greater distance on high-speed lines, or at a lesser distance on lower-speed and platform lines.

RS522 §1.1.3

28. Are AWS magnets provided at semaphore stop signals? What about a distant signal mounted on the same post as a stop signal?

Answer: No — AWS magnets are not provided at semaphore stop signals. However, where a distant signal is mounted on the same post as a semaphore stop signal, AWS is provided for the distant signal.

RS522 §1.1.3

29. Name the five types of AWS track magnet and state the purpose of each.

Answer: 1. Permanent magnet — triggers a warning if no energised electromagnet follows.
2. Electromagnet — when energised, gives a clear indication. 3. Suppressor magnet —

suppresses permanent magnets not applicable to a movement. 4. Depot test magnet — tests AWS at the exit of certain maintenance depots. 5. Portable magnet — gives a warning on the approach to temporary and emergency speed restrictions.

RS522 §1.2

30. What does the black (fully black) AWS visual indication mean?

Answer: It advises the driver that the associated signal is showing a green aspect or 'all clear'. It also means the audible warning has NOT been acknowledged — if not acknowledged, the brakes will be applied.

RS522 §1.3

31. What does the yellow and black AWS visual indication mean?

Answer: It advises the driver that a warning indication has been acknowledged.

RS522 §1.3

32. How long does the driver have to acknowledge an AWS audible warning? What happens if they don't?

Answer: The driver must acknowledge within two to three seconds. If not acknowledged, an emergency brake application will occur.

RS522 §1.3

33. What are the three states of the AWS isolation/fault indicator and what does each mean?

Answer: Off = AWS state is normal. Flashing = a fault has been detected in the train AWS equipment. On (steady) = the train AWS equipment has been isolated.

RS522 §1.3

34. List the aspects and indicators that will give the driver an AWS WARNING indication.

Answer: Colour light signal showing single or double yellow (steady or flashing) or red aspect; semaphore distant signal at caution; warning indicator for a permissible speed reduction; warning board for an ABCL, AOCL or open crossing; warning board or emergency indicator for a TSR or ESR; cancelling indicator for a warning not applicable to the train; AWS depot test magnet.

RS522 §1.4.1

35. Can AWS distinguish between a red, a double yellow, and a single yellow aspect?

Answer: No. AWS is not capable of distinguishing between a red, double yellow or single yellow aspect.

RS522 §1.4.1

36. When is a cancelling indicator provided, and what does it tell the driver?

Answer: A cancelling indicator is provided where the AWS magnet is not suppressed on a single or bi-directional line. It advises the driver that the AWS warning indication does not apply to trains travelling in that direction.

RS522 §1.6

37. Does AWS relieve the driver of the responsibility to observe and obey lineside signals and indicators?

Answer: No. AWS does not relieve the driver of the responsibility of observing and obeying lineside signals and indicators.

RS522 §1.1.2

Section 4 — TPWS (Train Protection & Warning System)

38. What does TPWS stand for?

Answer: Train Protection and Warning System.

RS522 §2.1

39. Is TPWS designed to prevent a Signal Passed at Danger (SPAD)?

Answer: No — it is designed to mitigate the consequences of a SPAD.

RS522 §2.1.2

40. At approximately what distance ahead of buffer stops is TPWS fitted, and at what speed does it activate?

Answer: Approximately 65 metres; at speeds greater than 10 mph.

RS522 §2.1.3

41. What are the two types of TPWS track equipment?

Answer: Train Stop System (TSS) and Overspeed Sensor System (OSS).

RS522 §2.2.1

42. When is the TSS loop energised?

Answer: When the signal is at danger.

RS522 §2.2.2

43. Must every signal fitted with TPWS have a TSS?

Answer: Yes — TSS is always present; it may also have one or more OSS.

RS522 §2.2.4

44. On the standard TPWS panel, what colour is the brake demand indicator?

Answer: Red.

RS522 §2.3.3

45. On a standard TPWS panel, how long after the brake demand was initiated will the brakes release?

Answer: 60 seconds after the brake demand was initiated.

RS522 §2.3.3

46. What is the Train Stop Override button and when is it used?

Answer: A yellow button used to override a TSS brake demand when authorised to pass a signal at danger.

RS522 §2.3.3

47. The enhanced TPWS panel has three separate brake demand indicator buttons. What are they?

Answer: SPAD (red), OVERSPEED (yellow), AWS (yellow).

RS522 §2.3.4

48. The TPWS temporary isolation switch — where is it mounted?

Answer: In the cab, but out of reach from the normal driving position.

RS522 §2.3.5

49. When would the full isolation switch need to be used?

Answer: When there is a fault with the AWS system, or the train has stopped over an AWS magnet and the driver cannot cancel it.

RS522 §2.3.6

50. TPWS becomes defective during a journey when it SHOULD be in operation. What must the driver do FIRST?

Answer: Stop the train immediately, then tell the signaller and await instructions.

GERT8000-TW5 §25.3

51. Permission is given for a passenger train to proceed with defective TPWS and no competent person is provided. What is the maximum speed?

Answer: Not exceeding 60 mph (95 km/h), or any lower permissible speed.

GERT8000-TW5 §25.5

52. On a Track Circuit Block (TCB) line, what must the signaller ensure when allowing a train with defective TPWS to proceed?

Answer: At least two controlled signals must be kept at danger between the defective train and any conflicting or converging movements ahead.

GERT8000-TW5 §25.5

Section 5 — TW5: Defective AWS / TPWS Procedures

53. When approaching a signal with a competent person present due to defective/isolated AWS or TPWS, what must the driver do?

Answer: Call out the signal aspect or indications and give a commentary on the speed reduction on the approach to cautionary and stop aspects.

GERT8000-TW5 §2.2

54. Can a train start a journey from a maintenance depot with AWS defective in a cab required to be driven from, when AWS is required to be in operation?

Answer: No.

GERT8000-TW5 §4.1

55. Without a competent person provided, what is the maximum speed for a freight train or OTM with defective/isolated AWS?

Answer: 50 mph (80 km/h).

GERT8000-TW5 §4.5

56. Without a competent person, all trains other than freight/OTM with defective AWS must not exceed what speed in poor visibility?

Answer: 40 mph (65 km/h).

GERT8000-TW5 §4.5

57. Under which circumstances may a driver isolate the AWS when it is required to be in operation?

Answer: When cancelling the AWS warning indication does not stop the horn sounding or brakes applying; when successive/intermittent failures suggest defective equipment; or when the train stops directly over the track equipment.

GERT8000-TW5 §4.4

58. If AWS was isolated because the train stopped with the receiver directly over the track equipment, what must the driver do when restarting?

Answer: Make sure the AWS is made operative again immediately, if possible.

GERT8000-TW5 §4.4

59. Is there an exception to the 'stop, tell the signaller and wait for instructions' requirement after isolating AWS?

Answer: Yes — if it is necessary to isolate AWS on an on-track machine before it can work in a protection zone.

GERT8000-TW5 §4.4

60. If a train fails to transition to ERTMS automatically as expected, and AWS and TPWS are operating, what may the driver do?

Answer: Proceed at normal permissible speed, obeying all signals.

GERT8000-TW5 §13.4

61. On a TCB line, or an ERTMS line with lineside signals, how many controlled signals must be kept at danger between a train with defective TPWS and any conflicting/converging movement ahead?

Answer: At least two.

GERT8000-TW5 §25.5(a)

62. On an absolute block (AB) line, when must the signaller accept a train with defective TPWS?

Answer: Not until the line is clear to the signaller's section signal.

GERT8000-TW5 §25.5(b)

63. On a non-TCB single line, what restriction applies to a train with defective TPWS approaching a crossing loop?

Answer: It must not approach if a train is approaching the crossing loop from the opposite direction.

GERT8000-TW5 §25.5(c)

64. If TPWS fails to activate when it should have done, what must the driver do?

Answer: Stop the train immediately, tell the signaller, and not move until instructed.

GERT8000-TW5 §25.4

Section 6 — Shunting, Permissive Working & Limit of Shunt

65. What is permissive working, and what authorises a train to use it?

Answer: It allows two trains to occupy one block section simultaneously, authorised by a subsidiary signal fitted on the same post as the stop signal.

Semaphore Signals Revision Notes, §6

66. When may a driver proceed on the authority of a subsidiary signal, and at what speed?

Answer: Only if the main stop signal aspect is not showing proceed; the driver must then proceed at caution.

Semaphore Signals Revision Notes, §6

67. Can a passenger train proceed on the authority of a semaphore shunting signal?

Answer: No — passenger trains must not proceed on the authority of a semaphore shunting signal. They may proceed on a subsidiary signal only when entering a permissive platform line.

GERT8000-S7 §3.1

68. A subsidiary signal clears, but the normal route indication is not shown. What must the driver do?

Answer: Proceed at caution, and be prepared to stop before any obstruction.

GERT8000-S7 §3.2

69. How should a driver read a row of shunting signal discs where three discs are displayed?

Answer: Top to bottom, left to right: the top disc applies to the left line, the middle disc to the middle line, and the bottom disc to the right line.

Semaphore Signals Revision Notes, §7

70. After a shunting movement made on the authority of a shunting or shunt-ahead signal, what two things must happen before continuing the journey?

Answer: The movement must return to the approach side of the signal, and the signal must display the appropriate proceed aspect/indication before continuing.

GERT8000-S7 §3.3

71. What is a "limit of shunt", and what is the consequence of passing it without permission?

Answer: A defined point beyond which a shunting movement must not pass without permission. Passing beyond it without permission is a SPAD (Signal Passed at Danger).

GERT8000-SS2 §5.5

72. Under what two conditions may a wrong-direction shunting movement go beyond a home signal on a falling gradient towards the next signal box?

Answer: a) the automatic brake is working throughout the train, or b) the locomotive is at the end nearer the next signal box.

GERT8000-SS2 §5.6

73. Who is "the shunter", and what must a driver do regarding their instructions?

Answer: A person competent to control specific shunting movements in and out of a depot, yard or sidings. The driver must work to the shunter's instructions even if a signal clears for the movement.

Classroom PPTX (Shunting), slide 35

74. Name at least four methods by which a shunting movement may be controlled.

Answer: Any four of: face to face, radio, hand signals, bell/buzzer, fixed shunting signals, fixed mainline signals.

Classroom PPTX (Shunting), slide 50

75. Before any shunting move begins, what must the driver and the person controlling the move clearly agree?

Answer: What exactly is to be done, and how the shunting movement will be controlled.

Classroom PPTX (Shunting), slide 40

76. At a stop board with no other signal present, what must a driver do?

Answer: Stop and wait for instruction.

Classroom PPTX (Shunting), slide 33

77. What is the maximum speed in a maintenance depot, servicing point or stabling siding, unless otherwise authorised?

Answer: 5 mph.

Classroom PPTX (Shunting), slide 64

78. What is the maximum speed in a washer, wheel lathe, shed, or fuelling point (F&I)?

Answer: 3 mph.

Classroom PPTX (Shunting), slide 64

Section 7 — Cab Setup & Train Preparation

79. What is the difference between Full Preparation and Partial Preparation/mobilisation?

Answer: Full Preparation involves performing all required exterior, cab and interior checks. Partial Preparation/mobilisation requires just a basic check of traction, braking, communication and door systems.

Classroom PPTX (Shunting), slide 16

80. When is an "above solebar" or "station" preparation acceptable instead of a full preparation?

Answer: Where a train has been berthed in a platform and it is impossible to check the exterior equipment.

Classroom PPTX (Shunting), slide 16

81. Why must a driver treat any electrical system on a traction unit as "live" at all times?

Answer: Because of the range of electrical voltages used to power equipment on motive power, every electrical system should be considered live regardless of its apparent state.

Classroom PPTX (Shunting), slide 17

82. When climbing into a vehicle, what three things must a driver do?

Answer: Put their bag in first; use both handrails and maintain three points of contact climbing each step; and ensure they are safely inside before picking up their equipment.

Classroom PPTX (Shunting), slide 18

83. Before climbing down from a vehicle, what must a driver check first?

Answer: That it is safe to climb down and that there are no train movements on opposite or adjacent lines.

Classroom PPTX (Shunting), slide 19

84. When descending from a vehicle, how should a driver climb down, and what must they never do?

Answer: Climb down backwards using both handrails, finding each step and maintaining three points of contact. They must never jump from the vehicle or miss steps.

Classroom PPTX (Shunting), slide 20

85. List at least five categories that systematic train preparation must check.

Answer: Any five of: mechanical components; lights, wipers and warning horns; traction and braking systems; fluid levels; safety systems; communication systems; emergency equipment; cleanliness and condition of the train.

Classroom PPTX (Shunting), slide 21

86. The "Train Prep" philosophy is compared to which other industry's checklist approach, and where is it documented for traction?

Answer: An airline pilot's checklist; the equivalent is found in Traction Manuals.

Classroom PPTX (Shunting), slide 22

87. If items are found missing or defective during preparation, who must this be reported to?

Answer: The maintenance department, the depot supervisor, or the operations controller.

Classroom PPTX (Shunting), slide 28

88. When preparing a train, what is the minimum number of detonators that must be available in the cab?

Answer: At least 10.

GERT8000-TW1 §11

89. How many red flags must be available in a driving cab generally, and how does this differ on a multiple-unit train?

Answer: Two red flags generally; on a multiple-unit train, one red flag must be available in each cab.

GERT8000-TW1 §11

90. When entering a driving cab before starting a journey, what must the driver make sure the DRA (Driver Reminder Appliance) is?

Answer: Set.

GERT8000-TW1 §10.1

Section 8 — Coupling & Uncoupling

91. What must a driver always check before coupling?

Answer: That it is safe to do so.

Classroom PPTX (Shunting), slide 86

92. If a "non-multi" sticker is displayed on a unit, what must the driver NOT do?

Answer: Must not attempt to couple the units.

Classroom PPTX (Shunting), slide 87

93. If two units do NOT display a "non-multi" sticker, what does this indicate, subject to the other checks?

Answer: That the units are compatible and coupling may be attempted, subject to checking coupling-head alignment and any other required checks.

Classroom PPTX (Shunting), slide 87

94. What physical alignment check must be made before coupling?

Answer: Check that both coupling heads are in alignment.

Classroom PPTX (Shunting), slide 87

95. What additional test does certain traction require once coupling has taken place?

Answer: A brake continuity test.

Classroom PPTX (Shunting), slide 87

96. If not working under the authority of a main or shunting signal, what must be obtained before coupling?

Answer: Permission from the signaller.

Classroom PPTX (Shunting), slide 87

97. Why is personal safety a particular concern when coupling?

Answer: Cab designs can offer limited views of the immediate area around the coupling point.

Classroom PPTX (Shunting), slide 89

98. Name three of the main components of an automatic (BSI) coupler.

Answer: Any three of: the main air connection, the locking pin and gathering horn coupling piston, and the electrical box.

Classroom PPTX (Shunting), slide 90

99. Where may a driver uncouple a traction unit from another traction unit on a running line?

Answer: Only at a signal box, at a signal, or on a platform line.

Classroom PPTX (Shunting), slide 94

100. Before uncoupling at a location where this is not done routinely, what must the driver do?

Answer: Tell the signaller what movements need to be made.

Classroom PPTX (Shunting), slide 94